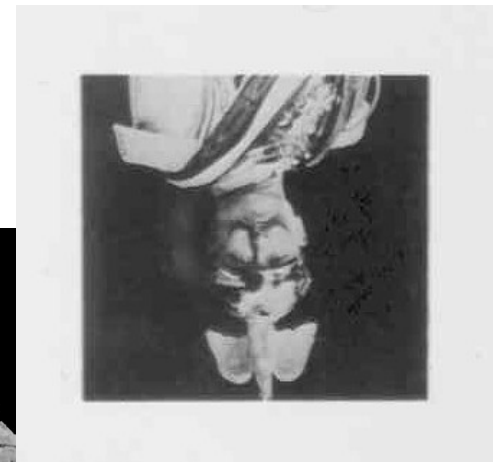
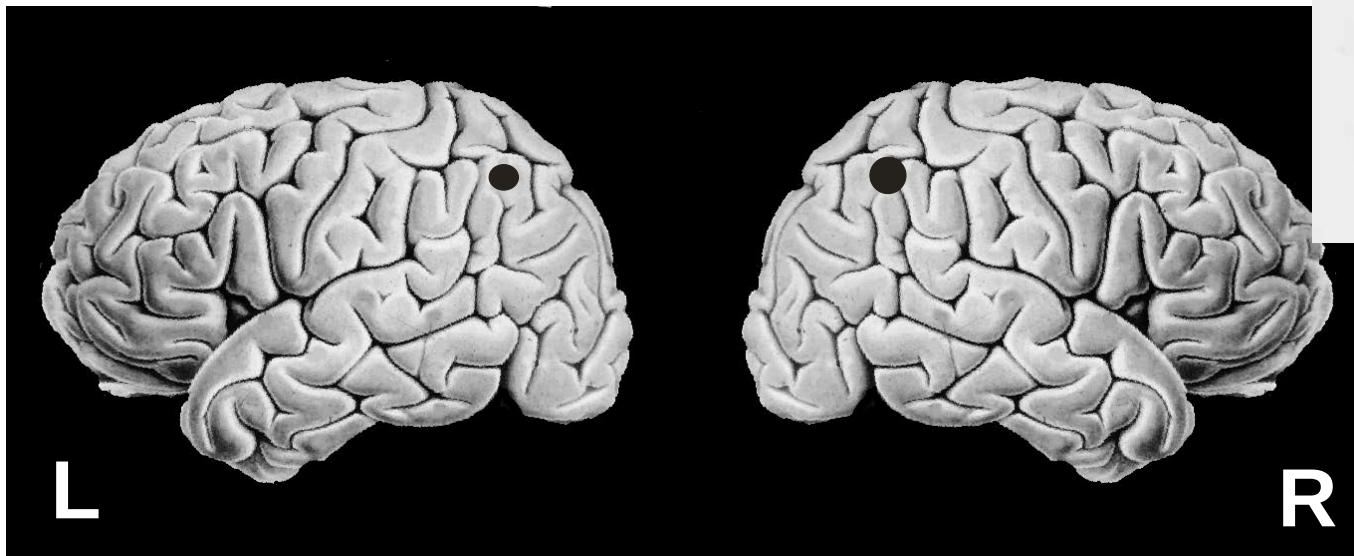


Object orientation agnosia

Object orientation agnosia

Patient Z. (Case 38)



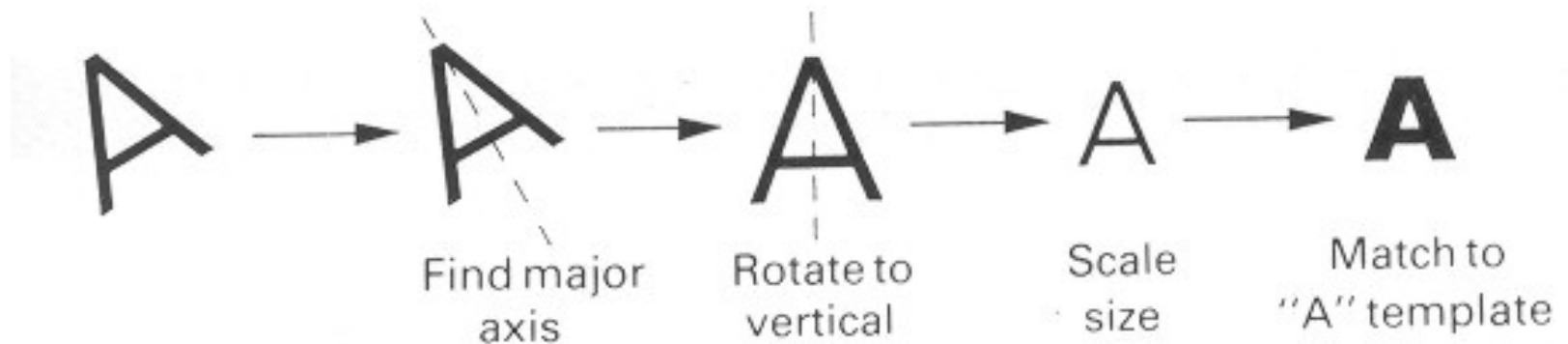
Ferber S, Karnath H-O (2003). Friedrich Best's case Z with misidentification of object orientation. In: C Code, C-W Wallesch, Y Joannette, AR Lecours (Hrsg.) *Classic Cases in Neuropsychology, Vol. II*. East Sussex, Psychology Press, 191-198.

Object recognition

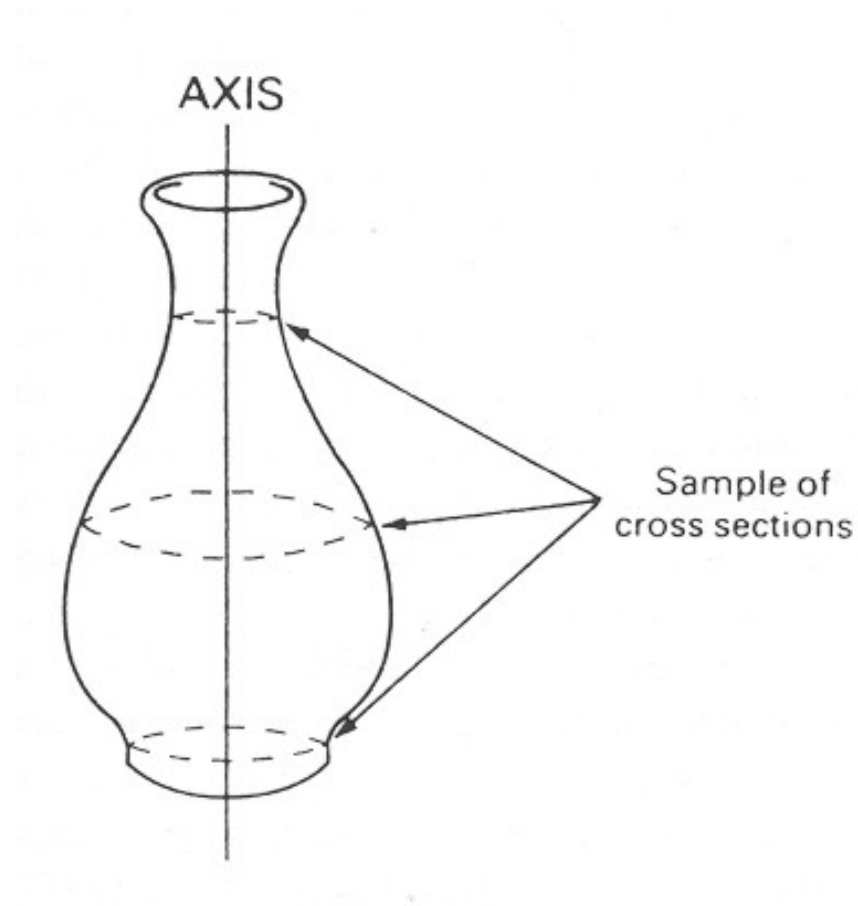
Viewpoint-dependent theories

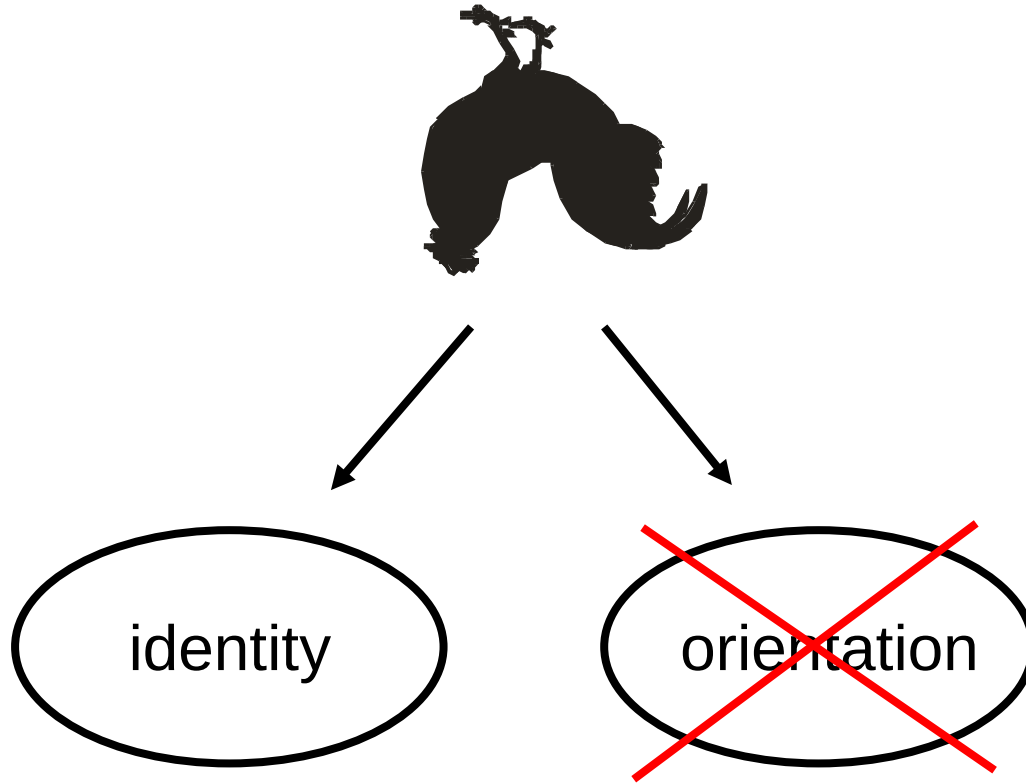
Viewpoint-invariant theories

Viewpoint-dependent theories



Viewpoint-invariant theories

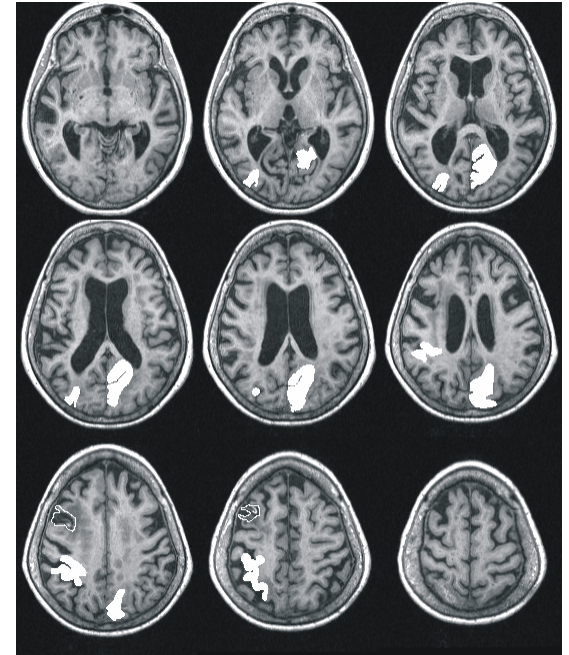


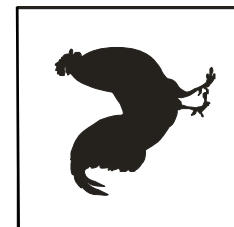
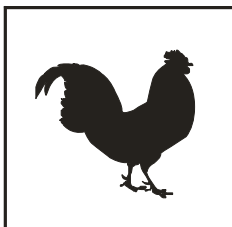
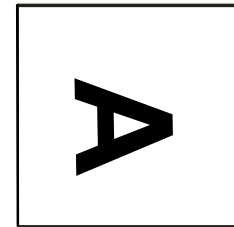
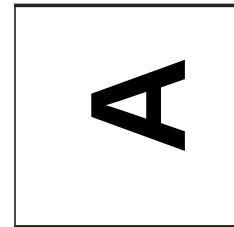
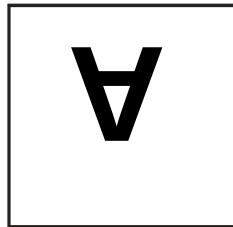
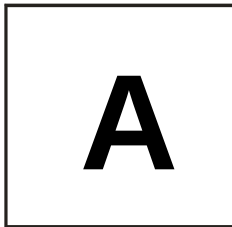


→ Object recognition without
viewer-centered information

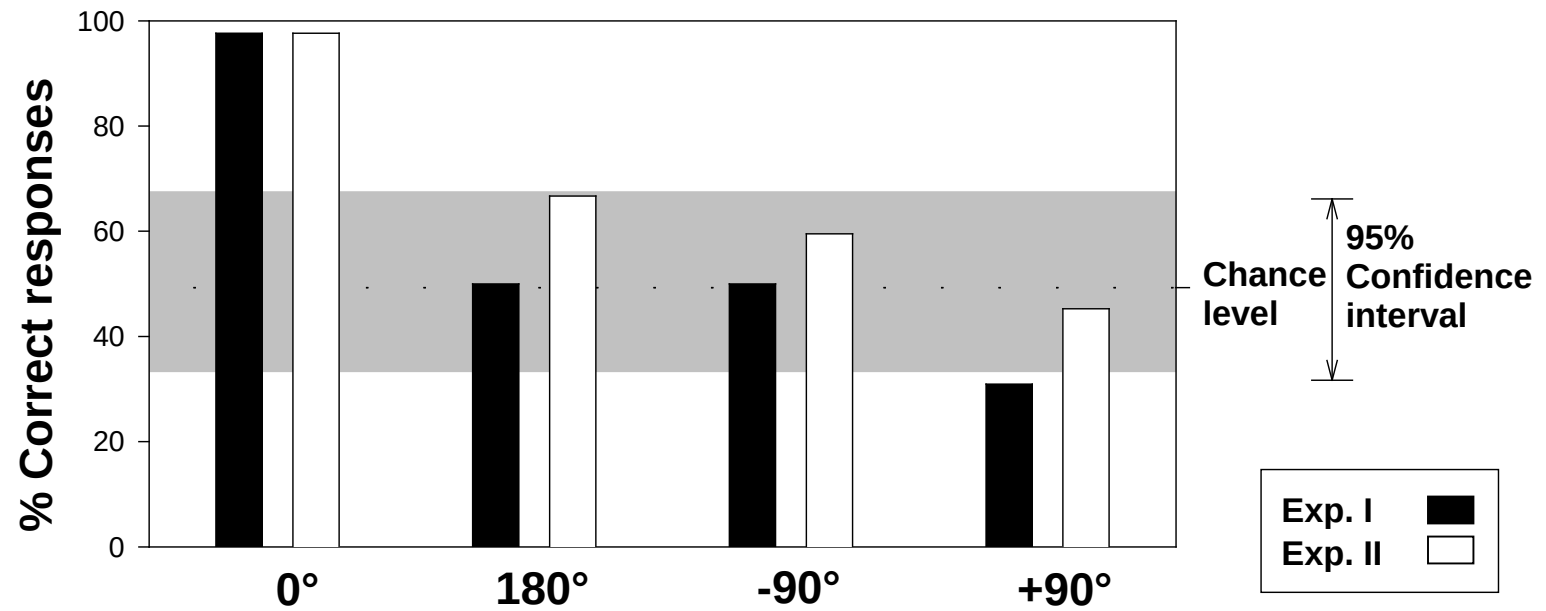
Patient KB

- 69-yr-old female
- 12-18 days post-stroke
- Bálint-Holmes syndrome:
simultanagnosia, spatial disorientation,
optic ataxia, impaired oculomotor behaviour
- no clinical signs of neglect or visual field defects





Letters



Objects are represented by neurons tuned to view, orientation, and size.

The number of tuned neurons for a particular orientation depends on the amount of experience with it.

Pusher syndrome

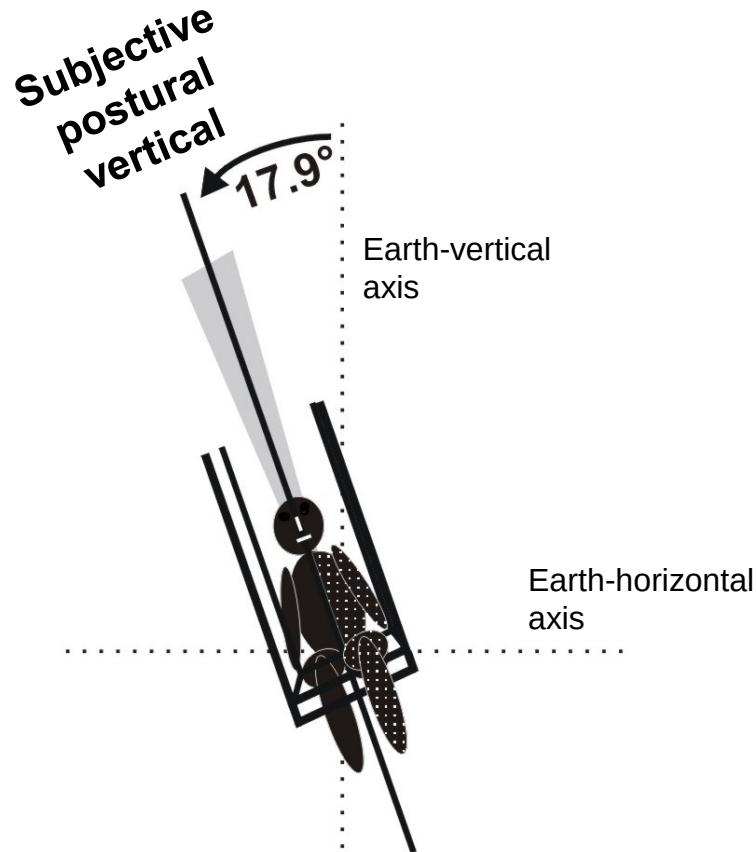
Contraversive pushing

(Davies, 1985)

A „pusher“ is a patient who uses the non-paretic extremities to actively push towards the side contralateral of the lesion (i.e., towards the hemiparetic side).

The patients resist any attempt to correct passively their tilted body posture.

Pusher Syndrome



Eyes closed

Contraverive pushing is due to an altered perception of the body's orientation in relation to gravity

Pusher patients experience their body as oriented 'upright' when actually tilted about 20° to the ipsilesional side

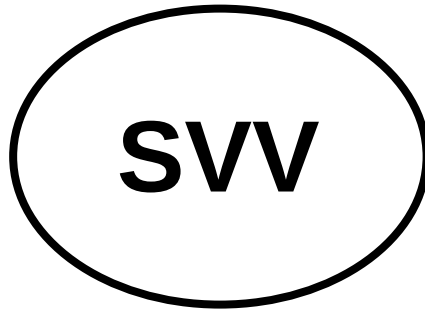
Subjective visual vertical (SVV)

Direction-specific indicator of a central and/or peripheral vestibular disorder

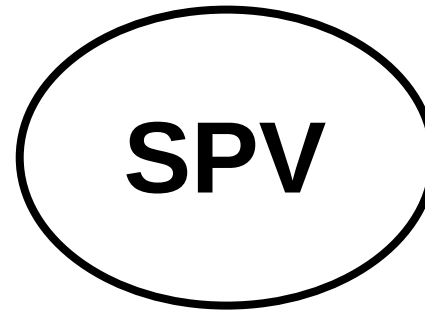
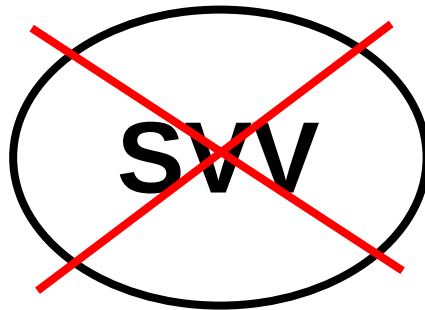
Subjective postural vertical (SPV)

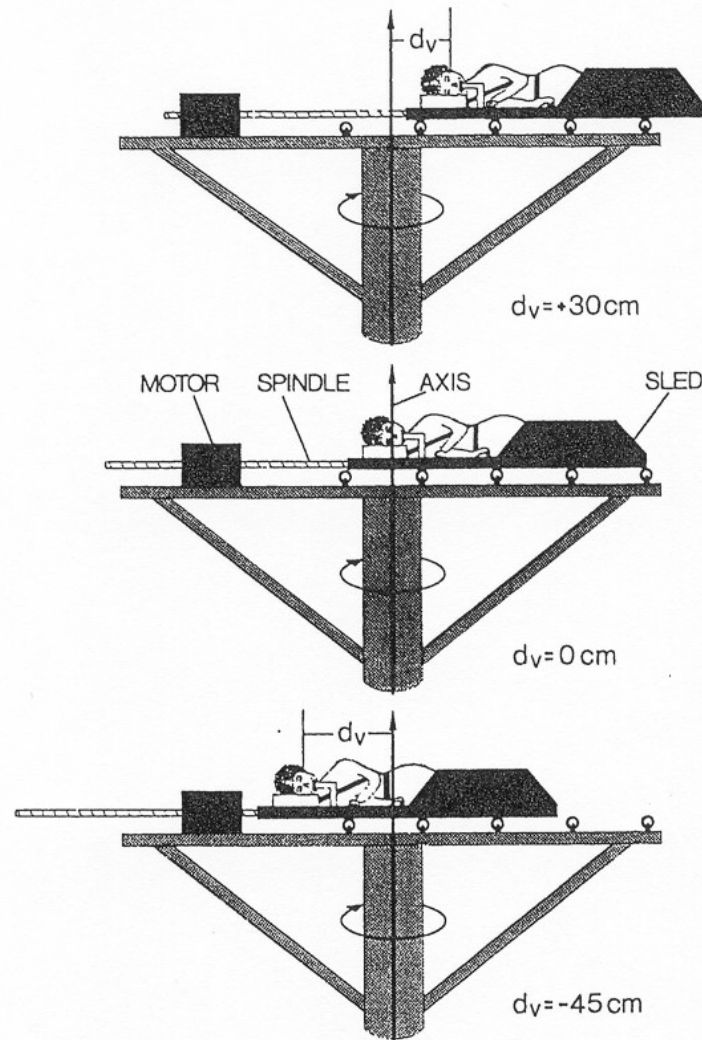
Reflects the subject's perception of earth-vertical body orientation

Pusher patients



Vestibular disorders



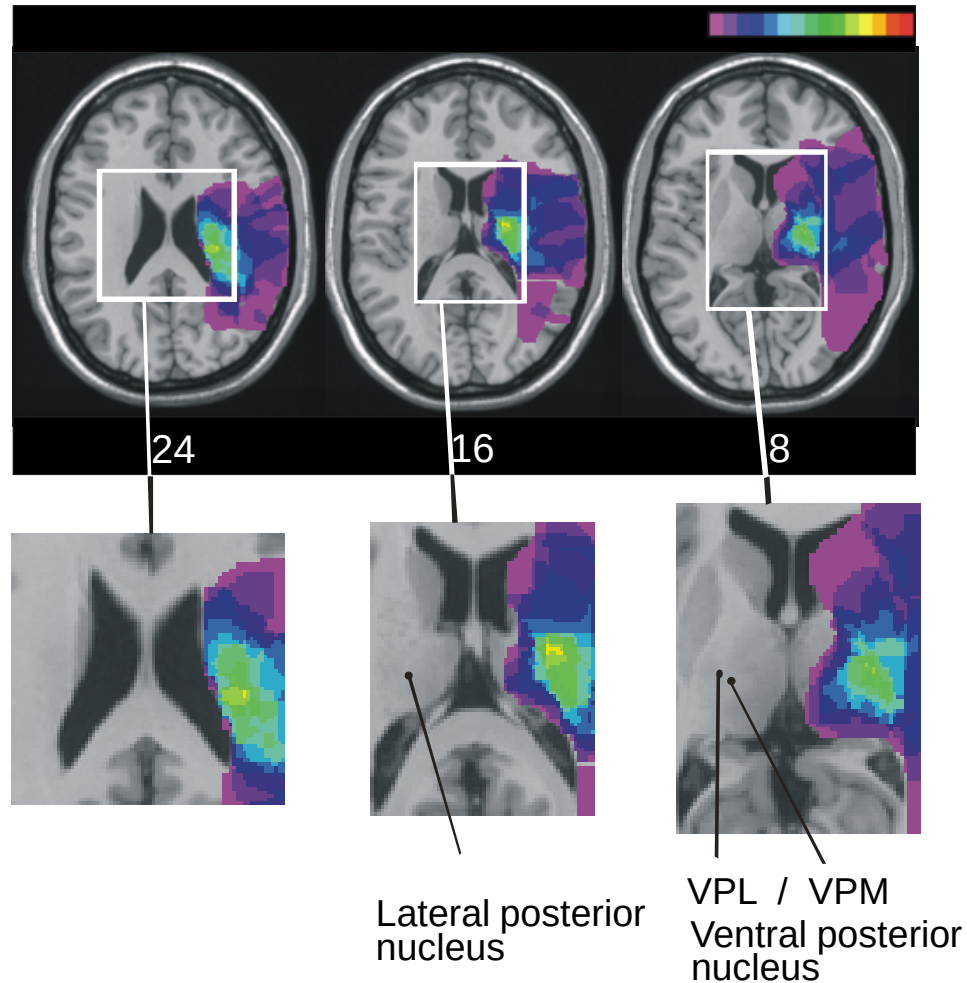


Mittelstaedt (1992, 1998)

Possible information source of the trunk-related graviceptive system:

- the inertia of a mass in the abdomen (blood in the large vessels and/or the mass of the abdominal viscera)

Lesion location in pusher syndrome



Lesion location in pusher syndrome

